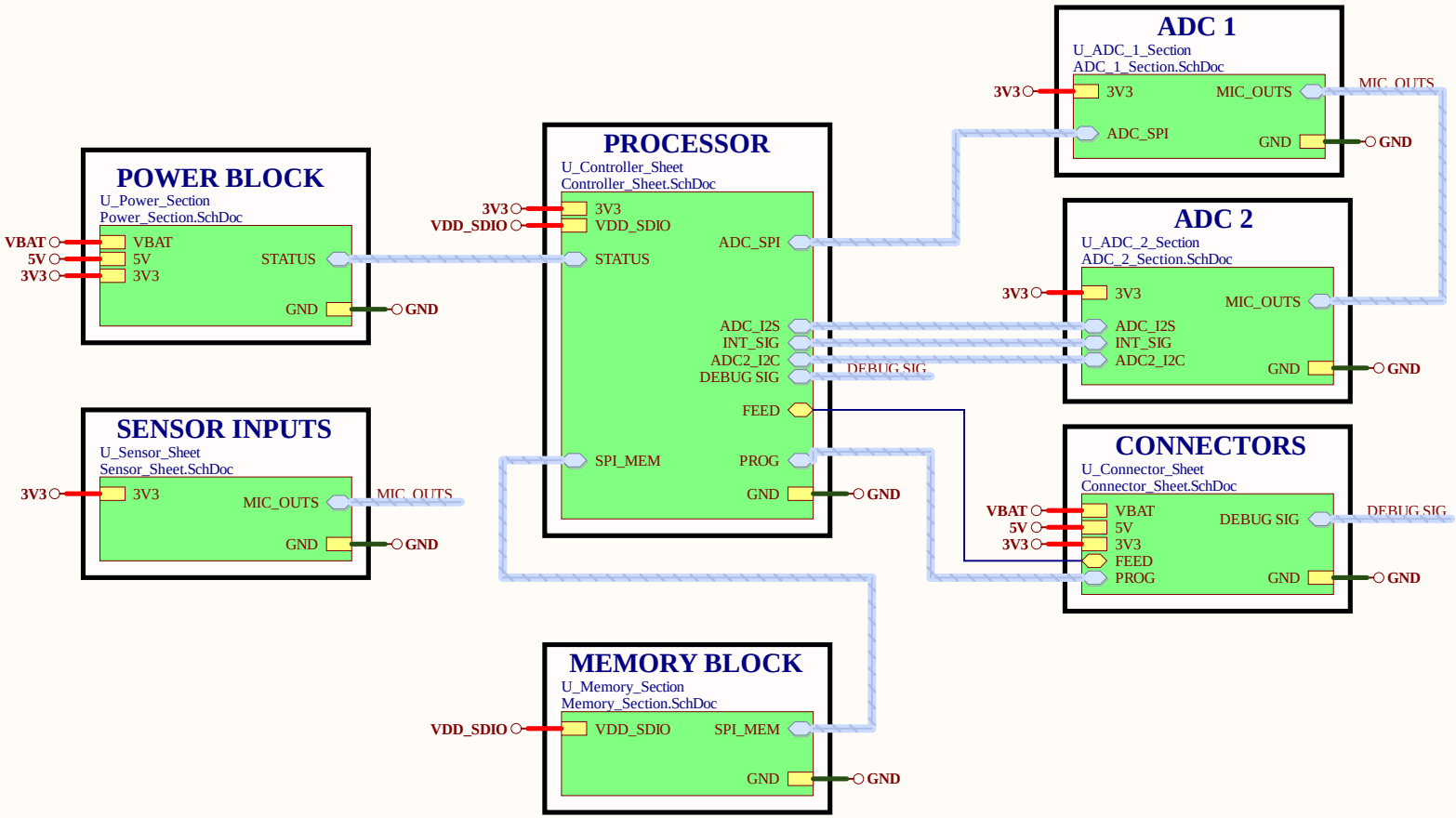
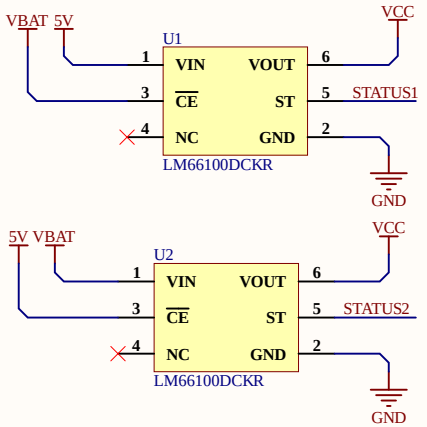


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	Size: A4	Number: 1	Revision: 1
	Date: 5/10/2022	Sheet 1 of 8	
	File: E:\Melih-SoundSensor\..\Top_Sheet.SchDoc	Drawn By: Huzaifa Ali	

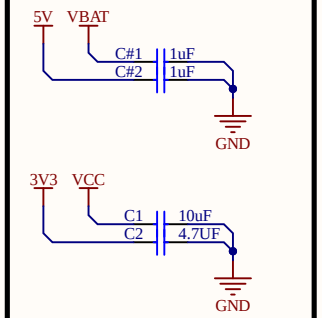
ORING STATUS SIGNAL



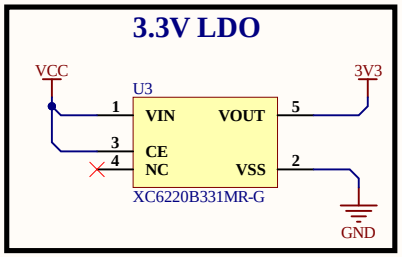
ORING CIRCUITRY



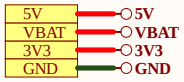
DECOUPLING CAPS



3.3V LDO

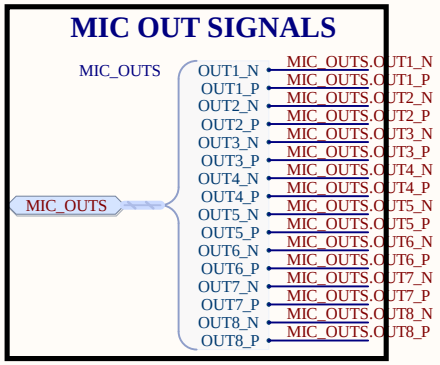


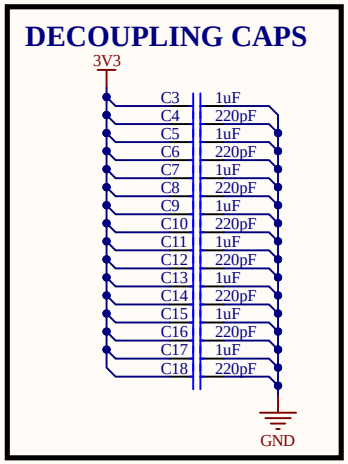
Fixed 3.3V output
150mV dropout @500mA
Max output current is 1A
Enable High to activitate the chip

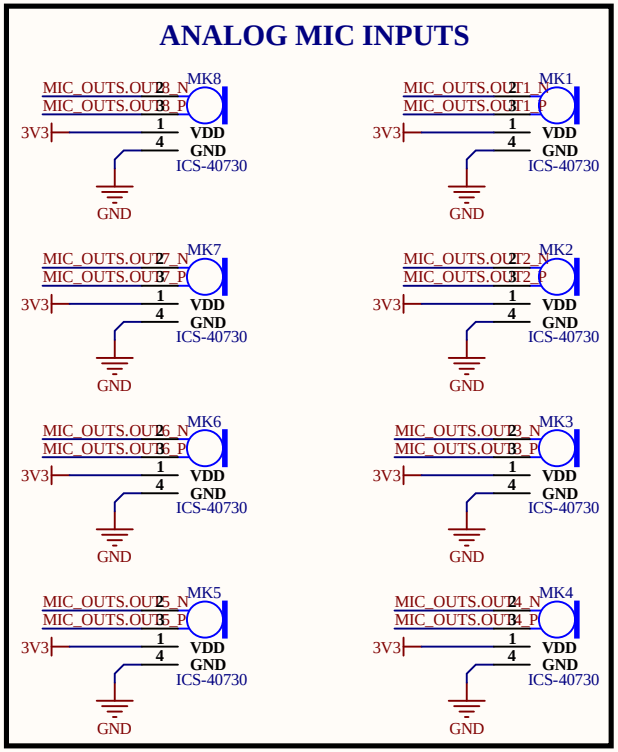


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Title: Sound_Sensor.PrjPcb			
Size: A4	Number: 1		Revision: 1
Date: 5/10/2022		Sheet 2 of 8	
File: E:\Melih-SoundSensor\..\Power_Section.SchDoc		Drawn By: Huzaifa Ali	

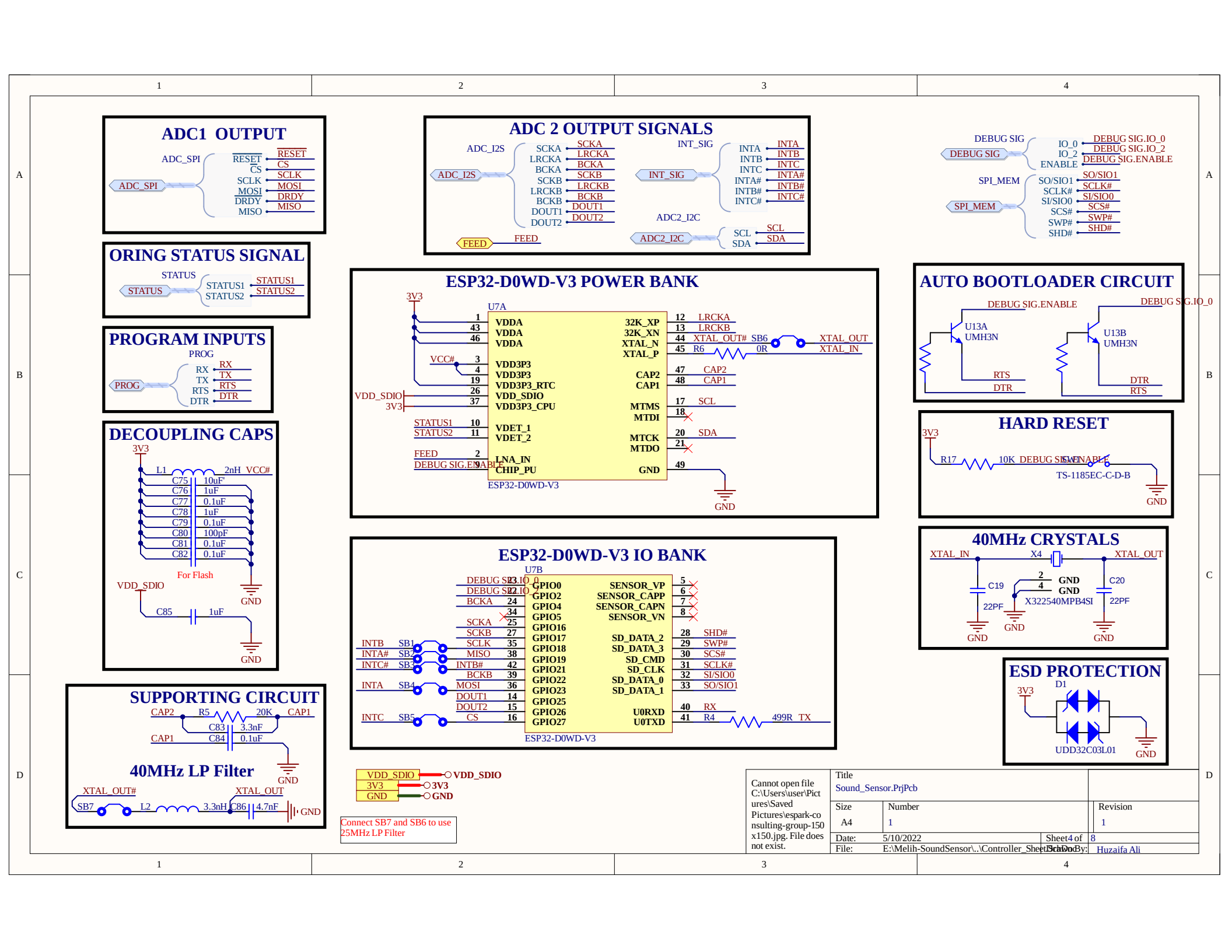


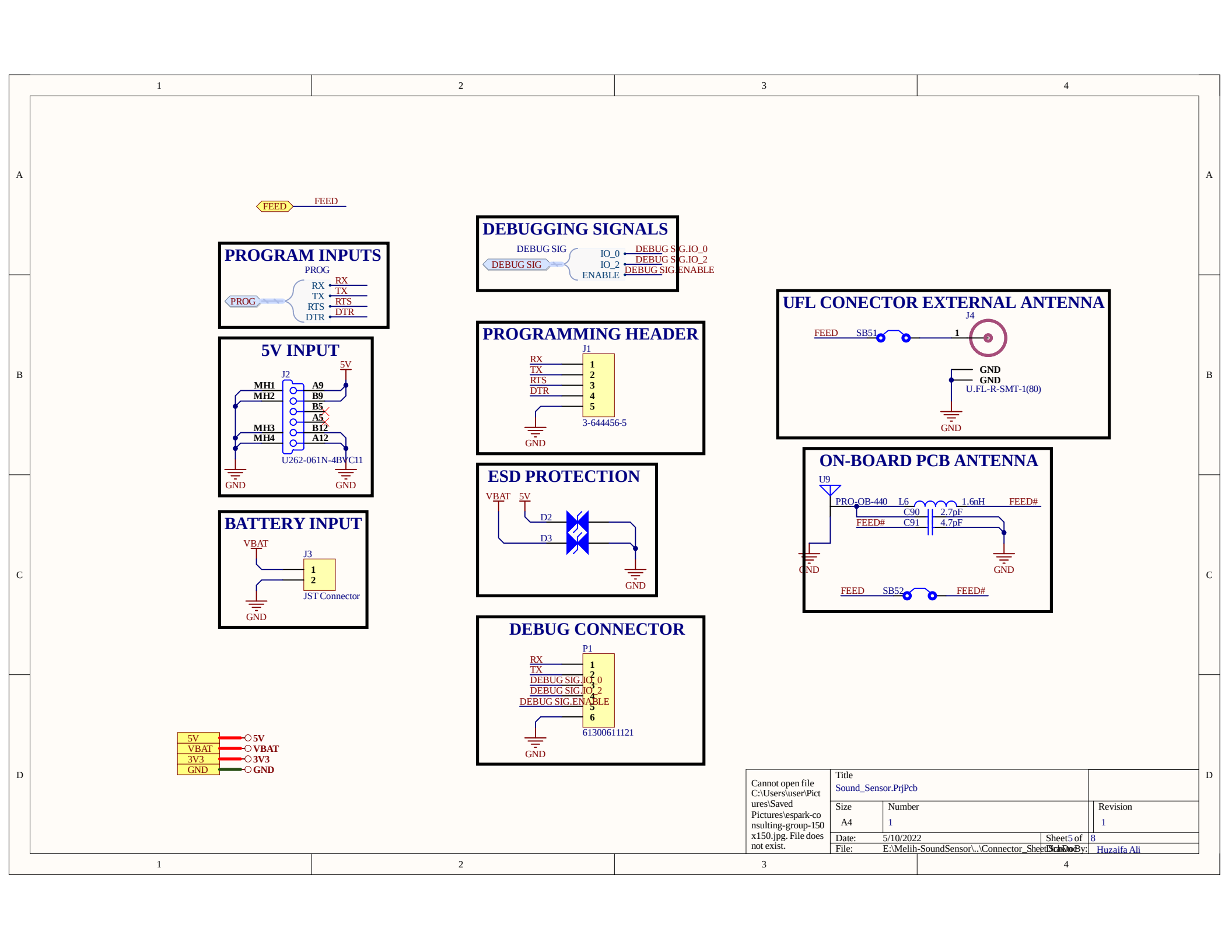




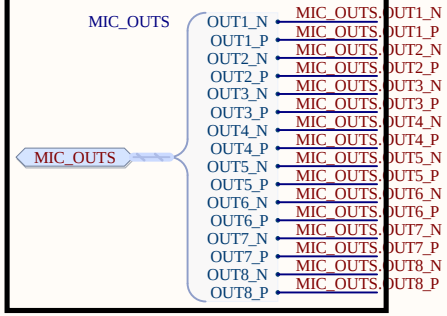


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	Size: A4	Number: 1	Revision: 1
	Date: 5/10/2022	Sheet 3 of 8	
	File: E:\Melih-SoundSensor\..\Sensor_Sheet.SchDoc	Drawn By: Huzaifa Ali	

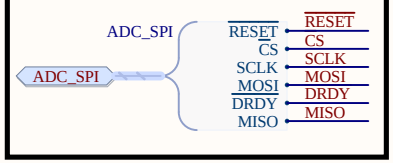

ADC1 OUTPUT
ADC_SPI
RESET
CS
SCLK
MOSI
DRDY
MISO
ADC 2 OUTPUT SIGNALS
ADC_I2S
SCKA
LRCKA
BCKA
SCKB
LRCKB
BCKB
DOUT1
DOUT2
FEED
INT_SIG
INTA
INTB
INTC
INTA#
INTB#
INTC#
ADC2_I2C
SCL
SDA
DEBUG SIG
IO_0
IO_2
ENABLE
DEBUG SIG.IO_0
DEBUG SIG.IO_2
DEBUG SIG.ENABLE
SPI_MEM
SO/SIO1
SCLK#
SI/SIO0
SCS#
SWP#
SHD#
ORING STATUS SIGNAL
STATUS
STATUS1
STATUS2
ESP32-D0WD-V3 POWER BANK
3V3
U7A
VDDA
VDD3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
VDET_1
VDET_2
LNA_IN
CHIP_PU
32K_XP
32K_XN
XTAL_N
XTAL_P
CAP2
CAP1
MTMS
MTDI
MTCK
MTDO
GND
LRCKA
LRCKB
XTAL_OUT#
XTAL_OUT
XTAL_IN
SB6
R6
0R
VCC#
SCL
SDA
ESP32-D0WD-V3
AUTO BOOTLOADER CIRCUIT
U13A
UMH3N
U13B
UMH3N
RTS
DTR
PROGRAM INPUTS
PROG
RX
TX
RTS
DTR
HARD RESET
R17
10K
SW-PB
TS-1185EC-C-D-B
DECOUPLING CAPS
L1
2nH
VCC#
C75
10uF'
C76
1uF
C77
0.1uF
C78
1uF
C79
0.1uF
C80
100pF
C81
0.1uF
C82
0.1uF
For Flash
C85
1uF
ESP32-D0WD-V3 IO BANK
U7B
GPIO0
GPIO2
GPIO4
GPIO5
GPIO16
GPIO17
GPIO18
GPIO19
GPIO21
GPIO22
GPIO23
GPIO25
GPIO26
GPIO27
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
SD_DATA_2
SD_DATA_3
SD_CMD
SD_CLK
SD_DATA_0
SD_DATA_1
U0RXD
U0TXD
SHD#
SWP#
SCS#
SCLK#
SI/SIO0
SO/SIO1
RX
TX
499R
R4
INTB
INTA#
INTC#
INTA
INTC
SB1
SB2
SB3
SB4
SB5
40MHz CRYSTALS
XTAL_IN
X4
XTAL_OUT
C19
C20
22PF
X322540MPB4SI
ESD PROTECTION
D1
UDD32C03L01
SUPPORTING CIRCUIT
CAP2
R5
20K
CAP1
C83
3.3nF
C84
0.1uF
40MHz LP Filter
XTAL_OUT#
XTAL_OUT
SB7
L2
3.3nH
C86
4.7nF
VDD_SDIO
3V3
GND
Connect SB7 and SB6 to use 25MHz LP Filter
Cannot open file C:\Users\user\Pictures\Saved Pictures\espark-consulting-group-150x150.jpg. File does not exist.
Title
Sound_Sensor.PrjPcb
Size
A4
Number
1
Revision
1
Date:
5/10/2022
Sheet 4 of 8
File:
E:\Melih-SoundSensor\..\Controller_Sheet.SchDoc
Drawn By:
Huzaifa Ali


FEED
FEED
PROGRAM INPUTS
PROG
RX
TX
RTS
DTR
5V INPUT
5V
J2
MH1
MH2
MH3
MH4
A9
B9
B5
A5
B12
A12
U262-061N-4BVC11
GND
GND
BATTERY INPUT
VBAT
J3
1
2
JST Connector
GND
5V
VBAT
3V3
GND
5V
VBAT
3V3
GND
DEBUGGING SIGNALS
DEBUG SIG
IO_0
IO_2
ENABLE
DEBUG SIG.IO_0
DEBUG SIG.IO_2
DEBUG SIG.ENABLE
PROGRAMMING HEADER
J1
3-644456-5
ESD PROTECTION
VBAT
5V
D2
D3
GND
DEBUG CONNECTOR
P1
DEBUG SIG.IO_0
DEBUG SIG.IO_2
DEBUG SIG.ENABLE
61300611121
UFL CONECTOR EXTERNAL ANTENNA
J4
SB51
GND
GND
U.FL-R-SMT-1(80)
ON-BOARD PCB ANTENNA
U9
PRO-OB-440
L6
1.6nH
FEED#
C90
2.7pF
C91
4.7pF
SB52
Cannot open file C:\Users\user\Pictures\Saved Pictures\espark-consulting-group-150x150.jpg. File does not exist.
Title
Sound_Sensor.PrjPcb
Size
A4
Number
1
Revision
1
Date:
5/10/2022
Sheet 5 of 8
File:
E:\Melih-SoundSensor\..\Connector_Sheet.SchDoc
Drawn By:
Huzaifa Ali

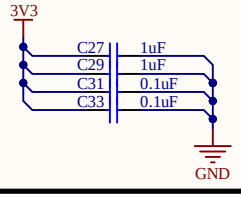
MIC OUT SIGNALS



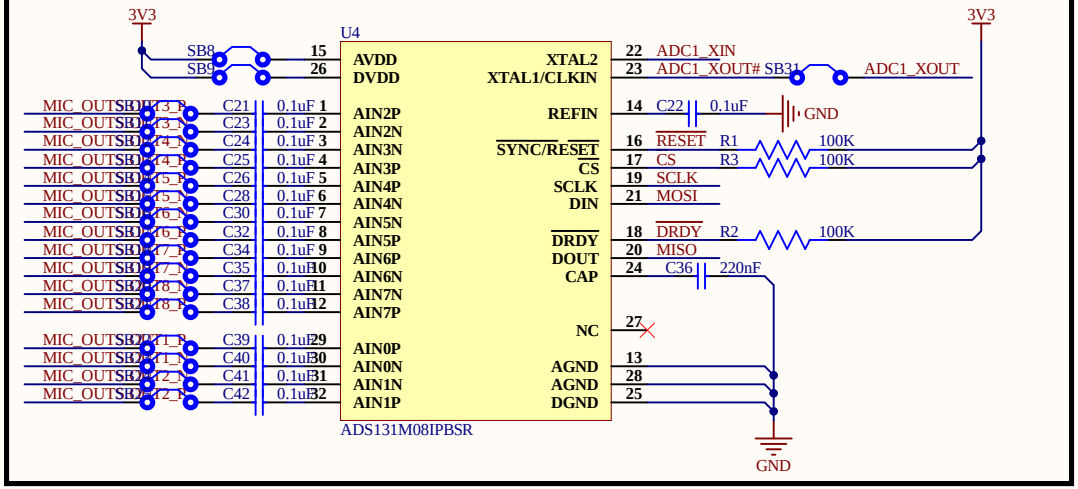
ADC OUTPUT



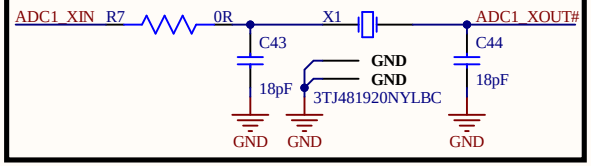
DECOUPLING CAPS



ADC WITH SPI INTERFACE

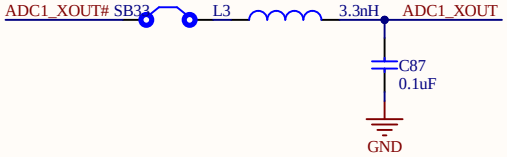


8.192 MHz Crystal





8.7MHz Low Pass Filter



Connect SB39 and SB31 to use 25MHz LP Filter

Title		
Size A4	Number	Revision
Date: 5/10/2022		Sheet of
File: E:\Melih-SoundSensor\..\ADC_1_Section.SchDoc		Drawn By:

I2C CONTROL SIGNALS

ADC2_I2C

ADC2_I2C — SCL — SCL
ADC2_I2C — SDA — SDA

MIC OUT SIGNALS

MIC_OUTS

MIC_OUTS — OUT1_N — MIC_OUTS.OUT1_N
MIC_OUTS — OUT1_P — MIC_OUTS.OUT1_P
MIC_OUTS — OUT2_N — MIC_OUTS.OUT2_N
MIC_OUTS — OUT2_P — MIC_OUTS.OUT2_P
MIC_OUTS — OUT3_N — MIC_OUTS.OUT3_N
MIC_OUTS — OUT3_P — MIC_OUTS.OUT3_P
MIC_OUTS — OUT4_N — MIC_OUTS.OUT4_N
MIC_OUTS — OUT4_P — MIC_OUTS.OUT4_P
MIC_OUTS — OUT5_N — MIC_OUTS.OUT5_N
MIC_OUTS — OUT5_P — MIC_OUTS.OUT5_P
MIC_OUTS — OUT6_N — MIC_OUTS.OUT6_N
MIC_OUTS — OUT6_P — MIC_OUTS.OUT6_P
MIC_OUTS — OUT7_N — MIC_OUTS.OUT7_N
MIC_OUTS — OUT7_P — MIC_OUTS.OUT7_P
MIC_OUTS — OUT8_N — MIC_OUTS.OUT8_N
MIC_OUTS — OUT8_P — MIC_OUTS.OUT8_P

DECOUPLING CAPS

3V3
C56 10uF
C57 10uF
C58 0.1uF
C59 0.1uF
GND

24.576 MHz Crystals

XI_A R8 0R X2 XO_A#
C61 18pF GND
C62 18pF GND
GND GND S3224576121060
GND GND GND

XI_B R9 0R X3 XO_B#
C73 18pF GND
C74 18pF GND
GND GND S3224576121060
GND GND GND

INT_SIG

INT_SIG — INTA — INTA
INT_SIG — INTB — INTB
INT_SIG — INTC — INTC
INT_SIG — INTA# — INTA#
INT_SIG — INTB# — INTB#
INT_SIG — INTC# — INTC#

ADC_I2S

ADC_I2S — SCKA — SCKA
ADC_I2S — LRCKA — LRCKA
ADC_I2S — BCKA — BCKA
ADC_I2S — SCKB — SCKB
ADC_I2S — LRCKB — LRCKB
ADC_I2S — BCKB — BCKB
ADC_I2S — DOUT1 — DOUT1
ADC_I2S — DOUT2 — DOUT2

ADC ICs

3V3
SB26 SB27 SB28 — 8 13 14 — AVDD DVDD IOVDD
MIC_OUTSB.OUT1_N … MIC_OUTSB.OUT4_P — C45 … C52 680nF — 1 2 3 4 27 28 29 30 — VINL2/VIN1M VINR2/VIN2M VINL1/VIN1P VINR1/VIN2P VINL4/VIN4M VINR4/VIN3M VINL3/VIN4P VINR3/VIN3P
C53 1uF C54 0.1uF — 11 — LDO
XI_A SCKA LRCKA BCKA — 10 15 16 17 — XI SCKI LRCK BCK
GND
U5
PCM1865DBTR
XO DOUT INT MD0 MD1 MD3 MD2 MD4 MD5 MD6 Mic Bias VREF AGND DGND
9 XO_A# SB39 XO_A
18 DOUT1
19 INTC
3V3
26 25 24 SCL 23 SDA 22 21 INTA 20 INTB
5
6 C55 1uF
7
12
GND

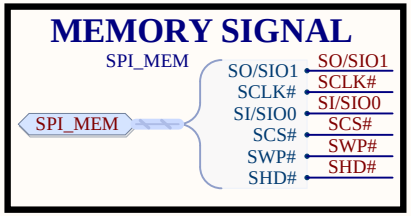
3V3
SB70 SB71 SB72 — 8 13 14 — AVDD DVDD IOVDD
MIC_OUTSB.OUT1_N … MIC_OUTSB.OUT4_P — C60 … C69 680nF — 1 2 3 4 27 28 29 30 — VINL2/VIN1M VINR2/VIN2M VINL1/VIN1P VINR1/VIN2P VINL4/VIN4M VINR4/VIN3M VINL3/VIN4P VINR3/VIN3P
C70 1uF C71 0.1uF — 11 — LDO
XI_B SCKB LRCKB BCKB — 10 15 16 17 — XI SCKI LRCK BCK
GND
U6
PCM1865DBTR
XO DOUT INT MD0 MD1 MD3 MD2 MD4 MD5 MD6 Mic Bias VREF AGND DGND
9 XO_B# SB49 XO_B
18 DOUT2
19 INTC#
26 25 24 SCL 23 SDA 22 21 INTA# 20 INTB#
GND
5
6 C72 1uF
7
12
GND

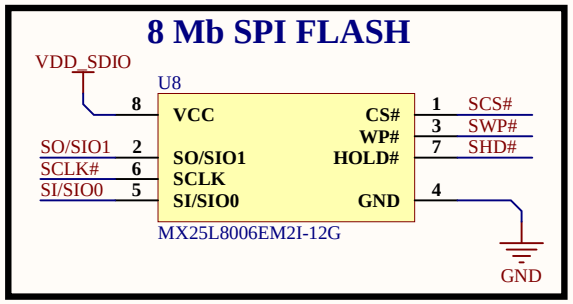
Connect SB39, SB49, SB50 and SB43 to use 25MHz LP Filter

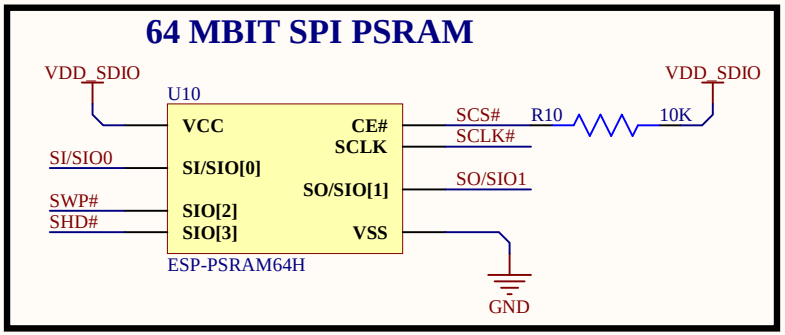
25 MHz Low Pass Filter

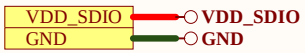
XO_A# SB43 L4 1.2nH XO_A
C88 33nF GND
XO_B# SB50 L5 1.2nH XO_B
C89 33nF GND

3V3 — 3V3
GND — GND

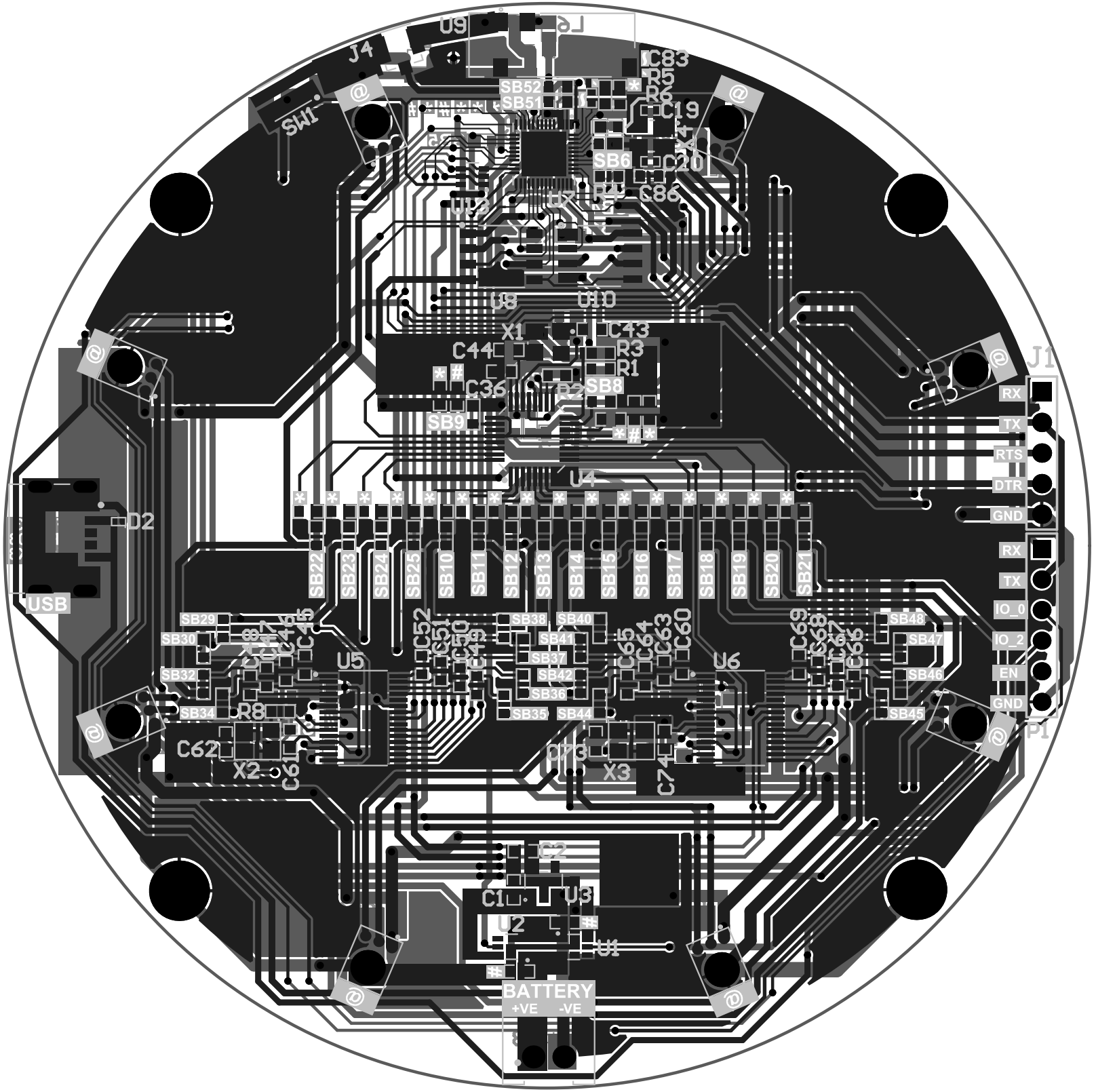








Title		
Size: A	Number	Revision
Date: 5/10/2022		Sheet of
File: E:\Melih-SoundSensor\..\Memory_Section.SchDoc		Drawn By:


J4
SW1
U9
IC83
R5
R6
C19
X4
SB52
SB51
SB6
C20
C86
U8
U10
X1
C43
C44
R3
R1
C36
R2
SB8
SB9
U4
J1
RX
TX
RTS
DTR
GND
RX
TX
IO_0
IO_2
EN
GND
PI
D2
USB
SB22
SB23
SB24
SB25
SB10
SB11
SB12
SB13
SB14
SB15
SB16
SB17
SB18
SB19
SB20
SB21
SB29
SB30
SB32
SB34
SB38
SB40
SB41
SB37
SB42
SB36
SB35
SB44
SB48
SB47
SB46
SB45
U5
U6
R8
C62
X2
X3
C2
C1
U3
U2
U1
BATTERY
+VE
-VE